Experiment-17

17. Windows OS Commands Execution (Traceroute, Ping, Ifconfig, Netstat)

Aim

To execute Windows OS networking commands to check network connectivity, trace the route to a destination, view IP configuration, and monitor active network connections.

Procedure

1. Open the Windows operating system.
2. Press `Win + R`, type `cmd`, and press Enter to open Command Prompt.
3. Execute the `tracert` command to trace the route to a destination.
4. Execute the `ping` command to check network connectivity.
5. Execute the `ipconfig` command to display IP configuration details.
6. Execute the `netstat` command to view active network connections.
7. Observe and record the outputs of all commands.

Code

Command 1: Traceroute

```
tracert google.com
```

Input 1:

```
C:\> tracert google.com
```

Output 1:

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SAIL>tracert google.com

Tracing route to google.com [142.250.183.174]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  172.23.16.1
  2    <1 ms    <1 ms    <1 ms  172.22.8.1
  3     2 ms    <1 ms     1 ms  14.139.187.225
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8     5 ms     9 ms     4 ms  lcmaaa-ar-in-f14.1e100.net [142.250.183.174]

Trace complete.

C:\Users\SAIL>
```

Command 2: Ping

`ping google.com`

Input 2:

`C:\> ping google.com`

Output 2:

```
C:\Users\SAIL>ping google.com

Pinging google.com [142.251.221.174] with 32 bytes of data:
Reply from 142.251.221.174: bytes=32 time=3ms TTL=117
Reply from 142.251.221.174: bytes=32 time=4ms TTL=117
Reply from 142.251.221.174: bytes=32 time=3ms TTL=117
Reply from 142.251.221.174: bytes=32 time=5ms TTL=117

Ping statistics for 142.251.221.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 5ms, Average = 3ms
```

Command 3: IP Configuration (Ifconfig equivalent)

Windows uses `ipconfig` instead of `ifconfig`.

`ipconfig`

Input 3:

```
C:\> ipconfig
```

Output 3:

```
C:\Users\SAIL>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c2b4:a84d:47c:7f56%15
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::e84:29b9:bc7d:a31a%10
   IPv4 Address. . . . . . . . . . . : 172.23.161.114
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . : 172.23.16.1

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6afa:a881:bf41:34dd%14
   IPv4 Address. . . . . . . . . . . : 192.168.72.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2efb:5d40:7ec7:43b5%6
   IPv4 Address. . . . . . . . . . . : 192.168.137.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

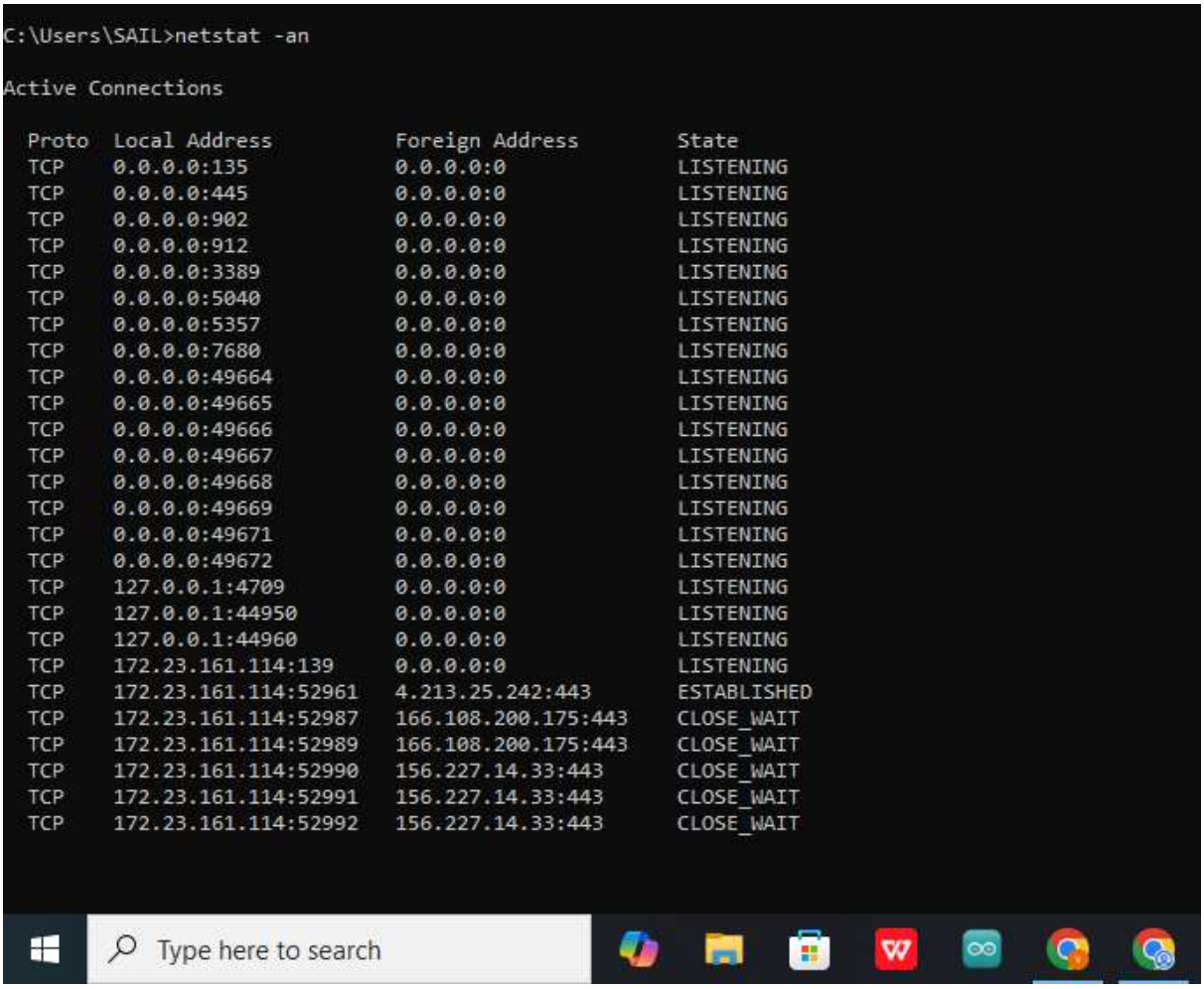
Command 4: Netstat

```
netstat -an
```

Input 4:

```
C:\> netstat -an
```

Output 4:



Result

The Windows OS networking commands `tracert`, `ping`, `ipconfig`, and `netstat` were studied and executed to trace network routes, test connectivity, display IP configuration, and monitor active network connections.